

Gamma Constants and Scaling Limits for Rounding Extinction

Cloitre's power-rounding conjecture, smooth-weight universality,
and the limits of regular variation

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Abstract

For $m > 0$, start from an integer $n \geq 0$ and repeatedly replace the current value by the largest multiple of k^m below it, at stages $k = 2, 3, \dots$. We prove that the first zero occurs at

$$\tau_m(n) \sim \left[m \Gamma\left(\frac{m}{m+1}\right)^{m+1} n \right]^{1/(m+1)}.$$

This establishes the real-parameter asymptotic conjecture recorded by Benoit Cloitre in OEIS A082528 and identifies the constants conjectured numerically for A082527 and A082528. The case $m = 1$ recovers the classical Tchoukaillon asymptotic and is not a new claim. The proof reverses the rounding process, establishes a deterministic scaling limit despite a discontinuous ceiling function, and solves the limit through an explicit Gamma product. We also prove the same limit for positive weights satisfying $k(w_k/w_{k-1} - 1) \rightarrow m$, give elementary finite-product certificates for the constant, and show by an exact block construction that ordinary regular variation alone does not determine it. Exact integer computations, including rational exponents, accompany the proof. Quantitative error terms, the bounded-error conjecture in the linear case, and more general microscopic rounding schedules remain open within this report.

Status. Complete mathematical proofs are supplied here; the report is AI-assisted, unrefereed, and not formally verified. The historical search was targeted, not exhaustive.

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1 The problem and the principal results

Repeated rounding is a simple deterministic operation with unexpectedly structured large-scale behavior. Fix a real number $m > 0$ and an integer $n \geq 0$. Define

$$x_1 = n, \quad x_k = k^m \left\lfloor \frac{x_{k-1}}{k^m} \right\rfloor \quad (k \geq 2), \quad \tau_m(n) = \min\{k \geq 1 : x_k = 0\}. \quad (1.1)$$

The value x_k is nonnegative and does not increase with k . The stopping time is finite: once $k^m > n$, the next rounding gives zero. We use the convention $\tau_m(0) = 1$. For $n > 0$, one always has $\tau_m(n) \geq 2$. When m is not an integer, the intermediate values can be real; the quotient x_k/k^m is nevertheless an integer for every $k \geq 2$.

The sequences for $m = 1, 2, 3$ are respectively [A073047](#), [A082527](#), and [A082528](#). Cloitre's entry for the cubic case also asks for the asymptotic law for every real $m > 0$ [[1](#), [2](#), [3](#)]. The following theorem determines that law and its constant.

Theorem 1.1 (Power-rounding extinction). *For every fixed real $m > 0$, set*

$$c_m = m \Gamma\left(\frac{m}{m+1}\right)^{m+1}, \quad L_m = \frac{1}{c_m}. \quad (1.2)$$

Then, as the integer n tends to infinity,

$$\tau_m(n) \sim (c_m n)^{1/(m+1)}. \quad (1.3)$$

Equivalently, if $T_m(K)$ is the smallest integer initial value that survives through stage K , then

$$T_m(K) \sim L_m K^{m+1}. \quad (1.4)$$

In particular,

$$\tau_2(n) \sim \left(2\Gamma(2/3)^3 n\right)^{1/3}, \quad (1.5)$$

$$\tau_3(n) \sim \left(3\Gamma(3/4)^4 n\right)^{1/4}. \quad (1.6)$$

The constants are

$$c_2 = 4.965917162430455473\dots, \quad c_3 = 6.764822981008760234\dots \quad (1.7)$$

These identify the values 4.96... and 6.76... conjectured numerically in the two OEIS entries. The specialization $c_1 = \pi$ is the classical Tchoukaillon constant, whose prior history is discussed in [section 2](#).

The same proof identifies more than a stopping index. Define the crossing times

$$t_0 = 1, \quad t_j = \prod_{r=1}^j \left(1 - \frac{1}{(m+1)r}\right) \quad (j \geq 1), \quad (1.8)$$

and the continuous function $y_m : (0, 1] \rightarrow [0, \infty)$ by

$$y_m(t) = \frac{(m+1)j-1}{m} t_{j-1} - jt, \quad t_j \leq t \leq t_{j-1}. \quad (1.9)$$

The definitions agree at each shared endpoint. Set

$$F_m(t) = c_m t^m y_m(t) \quad (0 < t \leq 1), \quad F_m(0) = 1. \quad (1.10)$$

Theorem 1.2 (The full extinction trajectory). *Let $J = \tau_m(n)$ for $n > 0$, and put*

$$X_n(t) = \frac{x_{\max(1, \lfloor Jt \rfloor)}}{n}, \quad 0 \leq t \leq 1.$$

Then

$$\sup_{0 \leq t \leq 1} |X_n(t) - F_m(t)| \longrightarrow 0. \quad (1.11)$$

Thus the fraction of the initial value remaining, plotted against the fraction of the extinction time elapsed, has an explicit deterministic limiting shape.

Our extension to more general weights uses a local condition on their successive ratios. This condition is stronger than specifying only their leading growth.

Theorem 1.3 (Smooth-weight universality). *Suppose $w_1 = 1$, $w_k > 0$, and*

$$k \left(\frac{w_k}{w_{k-1}} - 1 \right) \longrightarrow m > 0. \quad (1.12)$$

Replace k^m in [equation \(1.1\)](#) by w_k , and denote the corresponding threshold and stopping time by $T_w(K)$ and $\tau_w(n)$. Then

$$T_w(K) \sim L_m K w_K, \quad L_m \tau_w(n) w_{\tau_w(n)} \sim n. \quad (1.13)$$

With $J = \tau_w(n)$, the normalized forward trajectory converges uniformly to the same F_m in [equation \(1.10\)](#).

Theorem 1.3 allows logarithmic factors, shifts, and many small perturbations. Its hypothesis cannot simply be replaced by ordinary regular variation: [section 8](#) constructs weights $w_k \sim (k/r)^m$ for which the threshold constant is L_m/r .

1.1 What has and has not been resolved

Question or claim	Status in this report
A082528, every real $m > 0$	The stated leading-equivalence conjecture is proved; c_m is explicit.
A082527 and A082528 constants	The square and cube constants are determined by equations (1.5) and (1.6) .
A073047 leading equivalent	Recovered as a classical specialization, with no novelty claim.
A073047 bounded error	The stronger assertion $\tau_1(n) = \sqrt{\pi n} + O(1)$ is not proved here.
Full trajectories and smooth weights	Proved in theorems 1.2 and 1.3 .
Quantitative discrete errors	No rate is inferred from the compactness argument; these are explicit research questions.
Historical priority	No matching general power theorem was found in the targeted source search; exhaustive priority is not certified.

All limits involving m in the first three theorems hold for fixed $m > 0$. They make no assertion uniform in a moving parameter $m = m(n)$.

2 Historical context and the ProveIt starting point

2.1 The OEIS questions

Cloitre introduced the square and cube stopping sequences in April 2003. The entry for A082528 explicitly asks whether the first zero for real powers k^m has order $(c(m)n)^{1/(m+1)}$.

A082527 records the numerical constant $4.96\dots$, and A082528 records $6.76\dots$ [2, 3]. The entries were inspected during the source audit for this report. Their shared site-footer time is not treated as an individual sequence revision date.

The square entry contains a minor arithmetic slip in its example for $n = 10$: the correct intermediate value is $4\lfloor 10/4 \rfloor = 8$. The reported stopping index 3 is correct. The definitions, programs, and initial terms agree with [equation \(1.1\)](#); the slip has no effect on the theorem.

2.2 The classical linear sieve

When $m = 1$, the backward threshold is A002491. Starting with K , one successively rounds upward to a multiple of $K - 1, K - 2, \dots, 1$. Its first values are

$$1, 2, 4, 6, 10, 12, 18, 22, 30, 34, \dots$$

This is a standard description of the Tchoukaillon or Mancala sieve [4]. Erdős and Jabotinsky proved

$$T_1(K) = \frac{K^2}{\pi} + O(K^{4/3}) \quad (2.1)$$

and proposed the sharper $O(K)$ error [5]. Their paper already analyzes successive ranges of equal integer increments. The use of such ranges and a product limit in the present proof belongs to that established method.

There is a relevant caution about error terms. Broline and Loeb later claimed the sharper threshold estimate [6]. Knuth’s 2021 comment recorded in A002491 identifies an accumulation-of-errors problem in that argument and explains why its stated estimates do not establish the claimed improvement [4]. We use neither that sharpening nor any consequence of it. The classical [equation \(2.1\)](#) would imply

$$\tau_1(n) = \sqrt{\pi n} + O(n^{1/6}),$$

whereas the $O(1)$ error recorded as presumptive in A073047 is a stronger assertion.

Brown’s exposition obtains the ordinary constant from successive parabolic pieces and a Wallis product [7]. A 2023 answer by Thomas Lehéicy studies a different generalization, rounding upward to the r th admissible multiple, and obtains a ratio of Gamma functions [8]. Its use of fixed increment regimes and a final truncation estimate is methodologically relevant. Its parameter changes the rounding rule; the parameter m here changes the moduli themselves. The two statements concern different recurrences.

We also checked Li’s 2026 preprint on the two-dimensional Tchoukaillon array [9]. Its main theorem concerns a uniform quadratic approximation to array entries. It does not state the real-power extinction theorem proved here. None of our proofs depends on that preprint or on its discussion of sharp edge estimates.

2.3 Repository audit and choice of target

The research began with the current ProveIt repository [11]. Targeted searches and directory inspection showed extensive recent work on OEIS asymptotics, including Bell-scale growth, derivative expansion supports, and balanced-word enumeration. Those established results were taken into account when choosing a question with a clear remaining mathematical claim.

The repository search was indexed at revision

ce37e13f4aa16819c87c3ccc611362d15758f2ac.

Exact searches for A082527, A082528, A073047, A002491, and Tchoukaillon returned no matching treatment. We also inspected the 59-entry `oeis-sequence-asymptotics` directory and the 58-entry `series-and-transseries` directory at that revision. This is a bounded nonduplication check; it is not a claim to have read every file, binary, branch, or historical version of the repository.

The present article extends the repository’s research program through a new target. It is mathematically self-contained and assumes none of its unrefereed reports as a theorem. The substantive contribution claimed here is the real-parameter power theorem, its explicit constant, and the trajectory and weight extensions. The classical linear case and the increment-regime method are credited to the preceding literature.

[write] **Added 3 October 2026, batch 85.** This report is a single manuscript, printed in full: manuscript 05 of batch 85 of ProveIt’s incoming-reports intake, delivered as the archive `Rounding-Extinction_OEIS.zip` (wrapper directory `Rounding-Extinction/`) in commit `317c1ce2e`, and placed in commit `713149ded` in the research-report collection, among the OEIS-sequence asymptotics reports, as the report `a082528-rounding-extinction`. Its main file `article.tex` and the `\input` files `body.tex`, `extensions.tex`, `computation.tex`, `questions.tex` and `bibliography.tex` keep their delivered names. The author line and the PDF metadata read “Research report prepared for Vladimir Reshetnikov with OpenAI”: the package names OpenAI as the tool that prepared it and names no human author.

Pin and repository claims. The revision quoted above, `ce37e13f4`, is the ProveIt commit the manuscript was written against. Its two directory counts are true at that commit. At placement the intake repeated the search over the whole tracked tree (all reports, the `0eis`, number-theory and combinatorics projects, and the Lean and Rocq developments) for A073047, A082527, A082528, A002491, Tchoukaillon, Mancala, Jabotinsky and Broline and found no treatment: this report continues no repository material and has no neighbouring report. No statement here is formalized in Lean or Rocq, and placement in the collection confers no formal status.

Credit for $m = 1$. The linear case, $c_1 = \pi$, $T_1(K) \sim K^2/\pi$ and $\tau_1(n) \sim \sqrt{\pi n}$, is classical: the Tchoukaillon (Mancala) sieve of A002491 and the theorem of Erdős and Jabotinsky [5]. The report claims nothing new for it, nor for the increment-regime method; what it claims is the case $m \neq 1$ (real $m > 0$), the constants of [equations \(1.5\) and \(1.6\)](#), and the extensions of [theorems 1.2 and 1.3](#) and [sections 6 and 8](#).

OEIS facts checked by the intake. On 3 October 2026 the intake read the entries on `oeis.org` and confirmed the manuscript’s statements about them: A082528 records Cloitre’s conjecture for real $m > 0$ and the constant $6.76\dots$; A082527 records $4.96\dots$, and its example for $n = 10$ indeed prints the intermediate value 2 where $4\lfloor 10/4 \rfloor = 8$ is meant (the stopping index 3 is right); and A002491 carries Knuth’s 2021 comment that the error terms of Broline and Loeb accumulate, so that their argument does not give the claimed $O(K)$ bound. An independent integer implementation written at intake reproduced $T_1(1), \dots, T_1(10)$ as the A002491 prefix and every fixture term of the three stopping sequences, and gave $c_m T_m(3000)/3000^{m+1} = 0.999998, 1.00048, 1.00052$ for $m = 1, 2, 3$. The intake did not search the external literature for an earlier theorem for general m ; the novelty statement remains the manuscript’s, bounded as it says by a targeted search.

Notation. The manuscript reuses some letters with local meanings. Every symbol is printed as delivered; the table fixes each meaning, with the tempting false readings.

symbol	meaning here (and what it is <i>not</i>)
y_K, y_m	y_K is the interpolated backward array at terminal stage K (4.1); y_m is the limiting profile for the exponent m (1.9). The curve y_3 of figure 2 is the $m = 3$ limit, not the array at $K = 3$.
$T_m(K), T_w(K)$	the threshold for the exponent m , or for the weight sequence w ; in $T_3(4) = 88$ the subscript is m and the argument K .
$q_k^{(K)}, Q_k, Q; q, v$	$q_k^{(K)}$ is the backward quotient array (definition 3.1, and (7.5) for weights); $Q_k = x_k/k^m$ (or x_k/w_k) are the forward quotients; Q in lemma 3.2 is any integer quotient. In section 9, $m = p/q$ and the exponent q in (9.2) is that denominator, not a quotient; the quotient there is $v = q_k$.
a, a_k	$a = m/(m+1)$ in sections 5 and 6; $a_k = (k/(k-1))^m - 1$ (3.4) is the relative increment at stage k , not a term of a sequence tending to a . A generic $a > 1$ appears in the proof of corollary 3.3 and in R_a of proposition 8.1, and constants $a < \lambda^{1/(m+1)} < b$ in the proof of corollary 7.3.
b_k, b	$b_k = w_k/w_{k-1} - 1$ is the weighted analogue of a_k (section 7); $b \in \mathbb{R}$ is the logarithmic exponent of corollary 7.2.
t, t_j	$t \in [0, 1]$ is the fraction of the horizon; t_j (1.8) are the crossing times, decreasing from $t_0 = 1$, not the grid points j/K .
$c_m, L_m, C(u)$	c_m is the extinction constant and $L_m = 1/c_m$ the threshold constant ($T_m(K) \sim L_m K^{m+1}$); $C(u) = \max\{1, \lceil u \rceil\}$ (4.5) is the positive ceiling function, not a constant.
f_m, F_m	$f_m = t^m y_m$ tends to L_m at 0; $F_m = c_m f_m$ tends to 1 there.
r	the block length in proposition 8.1 (and $r = 3$ in the checks of section 9); elsewhere a summation or product index, the integer $r \geq 2$ in the proof of lemma 4.1, and the integer q th root in section 9.
$\delta, \delta_u, \delta_m$	δ is a cut-off parameter; δ_u is the unit mass at u in (6.5); δ_m is an error exponent in Research question 1.
w_k, v_k	the weights of theorem 1.3; v_k are the comparison weights after proposition 8.1.
z, ρ_m, J	$z(t) = my(t)/t$, and a complex variable in the proof of proposition 6.3; ρ_m is the loss density of corollary 6.2, not a radius; $J = \tau_m(n)$ (or $\tau_w(n)$) is the stopping index of the trajectory at hand.

The writing step prefixed every label with `rex:` (in `body.tex`, `extensions.tex`, `computation.tex` and `questions.tex`, every `\cref`, `\ref` and `\eqref` updated) and added three dated [write] notes: this one, one in section 9 on the shipped layout, and one in section B on the OEIS draft. It also added cross-reference aliases to the preamble, so that a current L^AT_EX prints “lemma 3.2” rather than “theorem 3.2” as the delivered PDF did, and set the double hyphens of one command line in section 9 as hyphens. No statement, proof or number of the manuscript was altered.

3 Exact reversal: extinction as a survival threshold

The forward process has a floor at every step and an initial value that changes with the question being asked. Fixing a terminal stage instead gives a convenient triangular recurrence.

Definition 3.1 (Backward quotient array). For an integer $K \geq 1$, define

$$q_K^{(K)} = 1, \quad q_{k-1}^{(K)} = \left\lceil \left(\frac{k}{k-1} \right)^m q_k^{(K)} \right\rceil \quad (2 \leq k \leq K), \quad (3.1)$$

and put $T_m(K) = q_1^{(K)}$. We suppress the superscript K when the terminal stage is fixed.

Lemma 3.2 (Survival duality). For every integer $n \geq 0$ and $K \geq 1$,

$$\tau_m(n) > K \iff n \geq T_m(K). \quad (3.2)$$

More generally, a value $k^m Q$ at stage $k \leq K$, with Q a nonnegative integer, survives through stage K if and only if $Q \geq q_k^{(K)}$.

Proof. At stage K , survival means a positive multiple of K^m , so its smallest integer quotient is 1. Suppose the least quotient that survives from stage k has been identified as q_k . An input $(k-1)^m Q$ produces an output at least $k^m q_k$ exactly when

$$\left\lceil \frac{(k-1)^m Q}{k^m} \right\rceil \geq q_k \iff Q \geq \frac{k^m q_k}{(k-1)^m}.$$

The least admissible integer Q is the ceiling in [equation \(3.1\)](#). Backward induction reaches stage 1, where the initial value n is already an integer and $1^m = 1$. $\square$

Corollary 3.3 (Exact inverse and strict increase). *The sequence $T_m(K)$ is strictly increasing and unbounded. For $n > 0$,*

$$\tau_m(n) = 1 + \max\{K \geq 1 : T_m(K) \leq n\}. \quad (3.3)$$

Equivalently, $\tau_m(n) = J$ if and only if $T_m(J-1) \leq n < T_m(J)$.

Proof. For terminal stage $K+1$, the quotient at stage K is $\lceil ((K+1)/K)^m \rceil \geq 2$, whereas it is 1 for terminal stage K . Every map $Q \mapsto \lceil aQ \rceil$ with $a > 1$ is strictly increasing on the integers. Reversing further therefore gives $T_m(K+1) > T_m(K)$. Formula (3.3) follows from [lemma 3.2](#). $\square$

3.1 The two estimates that control all rounding errors

Write

$$a_k = \left(\frac{k}{k-1} \right)^m - 1 = \frac{m}{k} + O_m(k^{-2}). \quad (3.4)$$

Because q_k is an integer,

$$q_{k-1} - q_k = \lceil a_k q_k \rceil \geq 1. \quad (3.5)$$

Consequently,

$$q_k \geq K - k + 1. \quad (3.6)$$

The inequality remains significant after scaling, even though $q_K/K \rightarrow 0$.

Multiplying the ceiling inequality by $(k-1)^m$ gives the second estimate:

$$0 \leq (k-1)^m q_{k-1} - k^m q_k < (k-1)^m. \quad (3.7)$$

Telescoping in either direction yields

$$0 \leq T_m(K) - k^m q_k < \sum_{j=1}^{k-1} j^m, \quad (k \geq 2), \quad (3.8)$$

$$k^m q_k \leq K^m + \sum_{j=k}^{K-1} j^m \leq K^m + \frac{K^{m+1}}{m+1}. \quad (3.9)$$

For $k=1$, the difference in [equation \(3.8\)](#) is exactly zero. The last inequality follows by comparing the increasing function x^m with its integral over $[1, K]$.

[Equation \(3.8\)](#) is the key matching estimate: after a macroscopic part of the backward trajectory has been computed, the entire remaining part changes its weighted value by at most an explicit sum. No assumption on the distribution of fractional parts is involved.

Example 3.4 (A cubic threshold). For $m=3$ and $K=4$, the backward quotients are

$$q_4 = 1, \quad q_3 = 3, \quad q_2 = 11, \quad q_1 = 88.$$

Thus $T_3(4) = 88$. The initial value 88 rounds through 88, 88, 81, 64, 0, so its stopping time is 5. The value 87 rounds through 87, 80, 54, 0 and stops at 4. This illustrates the strict inequality $\tau_m(n) > K$ in the survival convention.

4 A rigorous scaling limit for the backward array

For fixed K , define y_K by linear interpolation through

$$y_K(k/K) = q_k^{(K)}/K, \quad 1 \leq k \leq K. \quad (4.1)$$

Its domain is $[1/K, 1]$. The limit will be considered on compact subintervals of $(0, 1]$; the singular endpoint 0 is handled afterward using [equation \(3.8\)](#).

Lemma 4.1 (Compactness away from zero). *For every $0 < \delta < 1$, the restrictions of y_K to $[\delta, 1]$ are uniformly bounded and uniformly Lipschitz for all sufficiently large K . Every sequence of terminal stages tending to infinity has a subsequence whose interpolants converge locally uniformly on $(0, 1]$.*

Proof. For $k \geq \delta K/2$, [equation \(3.9\)](#) gives

$$\frac{q_k}{K} \leq \left(\frac{K}{k}\right)^m \left(\frac{1}{K} + \frac{1}{m+1}\right),$$

which is bounded independently of K . On an interpolation interval $((k-1)/K, k/K)$, its slope is

$$y'_K(t) = q_k - q_{k-1} = -[a_k q_k]. \quad (4.2)$$

The expansion in [equation \(3.4\)](#), together with $q_k = O_\delta(K)$ and $k \geq \delta K/2$, bounds this slope uniformly. The small enlargement to $\delta/2$ handles the single grid interval containing δ . Arzelà–Ascoli on $[1/r, 1]$, followed by a diagonal subsequence over integers $r \geq 2$, gives the stated compactness. $\square$

4.1 Selecting the nonzero terminal solution

Let y be any subsequential limit. Since $y_K(1) = 1/K$,

$$y(1) = 0. \quad (4.3)$$

[Equation \(3.5\)](#) implies, including between grid points,

$$y(s) - y(t) \geq t - s \quad (0 < s < t \leq 1). \quad (4.4)$$

In particular, $y(t) \geq 1 - t > 0$ for $t < 1$.

This is essential. If one wrote only $y' = -[my/t]$ with $y(1) = 0$, the identically zero function would solve that terminal-value problem. The positive integer quotient $q_K = 1$ forces a drift of at least one at each reversed step and rules out that spurious solution. We express this selection by the positive ceiling

$$C(u) = \max\{1, [u]\}, \quad u \geq 0. \quad (4.5)$$

For positive u , it is the ordinary ceiling.

Lemma 4.2 (Passing through the ceiling discontinuities). *Every subsequential limit is locally absolutely continuous and satisfies*

$$y'(t) = -C\left(\frac{my(t)}{t}\right) \quad \text{for almost every } t \in (0, 1), \quad (4.6)$$

together with [equations \(4.3\) and \(4.4\)](#).

Proof. Local Lipschitz continuity has already been proved. Define $z(t) = my(t)/t$. For $0 < s < t \leq 1$, [equation \(4.4\)](#) gives

$$\frac{y(s)}{s} - \frac{y(t)}{t} \geq y(t) \left(\frac{1}{s} - \frac{1}{t} \right) + \frac{t-s}{s} > 0.$$

Thus z is continuous and strictly decreasing as t increases. It is bounded on $[\delta, 1]$, so it assumes an integer value at only finitely many points in that interval.

Away from these exceptional points and $t = 1$, let $k = \lceil Kt \rceil$. The ratio expansion in [equation \(3.4\)](#) and local uniform convergence show

$$a_k q_k \longrightarrow \frac{my(t)}{t} = z(t).$$

The ceiling is continuous at the noninteger limit, so its value converges to $C(z(t))$. Excluding also the countable union of grid points, [equation \(4.2\)](#) therefore converges to the claimed right-hand side. The slope bounds from [lemma 4.1](#) permit dominated convergence in

$$y_K(t) = \frac{1}{K} - \int_t^1 y'_K(s) ds.$$

It follows that

$$y(t) = \int_t^1 C\left(\frac{my(s)}{s}\right) ds,$$

which proves [equation \(4.6\)](#). □

The argument does not assume uniform convergence of ceiling functions. It proves convergence away from a finite exceptional set on each compact interval and then uses an integrable uniform bound. That distinction is what makes the limiting equation rigorous.

4.2 Solving the increment regimes

Proposition 4.3 (Explicit profile and uniqueness). *The function satisfying [equations \(4.3\), \(4.4\)](#) and [\(4.6\)](#) is unique. It is precisely y_m in [equation \(1.9\)](#); consequently*

$$y_K \longrightarrow y_m \quad \text{uniformly on every } [\delta, 1], \quad \delta > 0. \tag{4.7}$$

Proof. The function $z(t) = my(t)/t$ decreases strictly from infinity to zero as t increases from zero to one. Its divergence at zero follows from $y(t) \geq 1 - t$. Therefore every positive integer j is attained at a unique crossing time t_j .

On (t_j, t_{j-1}) , one has $j-1 < z(t) < j$ and hence $y'(t) = -j$. Using $y(t_{j-1}) = (j-1)t_{j-1}/m$ gives

$$y(t) = \frac{j-1}{m} t_{j-1} + j(t_{j-1} - t).$$

At the next crossing, $y(t_j) = jt_j/m$, so

$$(m+1)j t_j = ((m+1)j - 1)t_{j-1}.$$

This is exactly the product recurrence in [equation \(1.8\)](#), and the formula for y is [equation \(1.9\)](#). Since $t_0 = 1$, there is no undetermined constant. Every subsequential limit is the same. If convergence of the whole family failed on some compact interval, a failing subsequence would have a further convergent subsequence by [lemma 4.1](#), contradicting this uniqueness. □

For example, the last portion of every limiting trajectory is $y_m(t) = 1 - t$ on $m/(m+1) \leq t \leq 1$. The next portion has slope -2 , the next slope -3 , and so on. These slopes describe the integer quotient changes; the weighted remaining value is $t^m y_m(t)$.

5 The Gamma constant and the extinction law

5.1 Evaluating the crossing product

Write $a = m/(m+1)$, so $0 < a < 1$. The Gamma functional equation gives

$$t_j = \frac{\Gamma(j+a)}{\Gamma(a)\Gamma(j+1)}. \quad (5.1)$$

The needed asymptotic is

$$t_j \sim \frac{j^{-1/(m+1)}}{\Gamma(m/(m+1))}. \quad (5.2)$$

For completeness, this can be proved without invoking an asymptotic expansion for Γ . The beta integral satisfies

$$B(a, j) = \int_0^1 u^{a-1}(1-u)^{j-1} du = \frac{\Gamma(a)\Gamma(j)}{\Gamma(j+a)}, \quad t_j = \frac{1}{jB(a, j)}.$$

After the substitution $v = ju$,

$$j^a B(a, j) = \int_0^j v^{a-1}(1-v/j)^{j-1} dv \rightarrow \Gamma(a).$$

Indeed, for $j \geq 2$ the integrand, extended by zero to $[j, \infty)$, is bounded by $v^{a-1}e^{-v/2}$ and converges pointwise to $v^{a-1}e^{-v}$. Dominated convergence proves the limit and [equation \(5.2\)](#).

Lemma 5.1 (The small-time weighted limit). *The function $f_m(t) = t^m y_m(t)$ has a finite limit at zero, and*

$$\lim_{t \downarrow 0} f_m(t) = \frac{1}{m\Gamma(m/(m+1))^{m+1}} = L_m. \quad (5.3)$$

Moreover,

$$-t^m \leq f'_m(t) \leq 0 \quad \text{almost everywhere.} \quad (5.4)$$

Proof. Writing $z = my_m/t$ and applying [equation \(4.6\)](#),

$$f'_m(t) = t^m(z(t) - C(z(t))).$$

For $z \geq 0$, the difference belongs to $[-1, 0]$. This proves [equation \(5.4\)](#). Since t^m is integrable on $(0, 1)$, f_m has a finite limit at zero. At a crossing time,

$$f_m(t_j) = \frac{j}{m} t_j^{m+1} \rightarrow \frac{1}{m\Gamma(m/(m+1))^{m+1}},$$

by [equation \(5.2\)](#). These crossing times tend to zero, so their limit is the full limit of f_m . $\square$

5.2 Matching to the unscaled initial value

Proof of theorem 1.1. Fix $0 < \delta < 1$ and set $k = \lfloor \delta K \rfloor$. The backward profile theorem implies

$$\frac{k^m q_k}{K^{m+1}} = \left(\frac{k}{K}\right)^m \frac{q_k}{K} \rightarrow f_m(\delta).$$

The elementary Riemann sum gives

$$\frac{1}{K^{m+1}} \sum_{j=1}^{k-1} j^m \rightarrow \frac{\delta^{m+1}}{m+1}.$$

Divide [equation \(3.8\)](#) by K^{m+1} to obtain

$$\begin{aligned} f_m(\delta) &\leq \liminf_{K \rightarrow \infty} \frac{T_m(K)}{K^{m+1}} \\ &\leq \limsup_{K \rightarrow \infty} \frac{T_m(K)}{K^{m+1}} \leq f_m(\delta) + \frac{\delta^{m+1}}{m+1}. \end{aligned}$$

Now let $\delta \downarrow 0$ and use [lemma 5.1](#). This proves [equation \(1.4\)](#). The two limits were taken in the specified order; no uniform approximation near the singular endpoint has been assumed.

For $J = \tau_m(n)$, the exact inverse gives $T_m(J-1) \leq n < T_m(J)$. As $n \rightarrow \infty$, necessarily $J \rightarrow \infty$, and both endpoints are asymptotic to $L_m J^{m+1}$. Thus $n \sim L_m J^{m+1}$, which is [equation \(1.3\)](#). $\square$

5.3 The whole forward trajectory

Proof of [theorem 1.2](#). Let $Q_k = x_k/k^m$ for $k \geq 2$, and set $Q_1 = n$. If $J = \tau_m(n)$, the forward trajectory survives stage $J-1$ but not stage J . The stagewise form of [lemma 3.2](#) therefore gives

$$q_k^{(J-1)} \leq Q_k < q_k^{(J)} \quad (1 \leq k \leq J-1). \quad (5.5)$$

Both bounding arrays have the same limiting shape, including after replacing the terminal scale $J-1$ by J . Their local uniform convergence and the continuity of y_m give

$$\frac{Q_{\lfloor Jt \rfloor}}{J} \rightarrow y_m(t) \quad \text{locally uniformly for } 0 < t < 1.$$

Combining this with $n \sim L_m J^{m+1}$ proves $X_n(t) \rightarrow c_m t^m y_m(t)$ for $0 < t < 1$. At the endpoints, $X_n(0) = 1$ and $X_n(1) = 0$. The limit F_m is continuous on $[0, 1]$ by [lemma 5.1](#).

To promote pointwise convergence to uniform convergence, choose a finite partition of $[0, 1]$ on whose subintervals the oscillation of F_m is less than a prescribed ε . For large n , convergence holds within ε at every partition point. Since each X_n is nonincreasing, its values between consecutive partition points lie between its endpoint values. The resulting error is at most 2ε throughout the interval. This proves [equation \(1.11\)](#). $\square$

The law describes the entire normalized loss curve, even at its early stages where the unweighted quotient $y_m(t)$ diverges. That endpoint is regular after multiplication by t^m .

6 Finite certificates and the dependence on the exponent

6.1 Finite products bracket the exact constant

The constant in [equation \(1.2\)](#) is expressed by a standard special function, but it also admits rigorous elementary enclosures.

Proposition 6.1 (Product certificates). *For every $m > 0$ and integer $j \geq 1$,*

$$\frac{j}{m} t_j^{m+1} \leq L_m \leq \left(\frac{j}{m} + \frac{1}{m+1} \right) t_j^{m+1}. \quad (6.1)$$

Consequently,

$$\frac{t_j^{-(m+1)}}{j/m + 1/(m+1)} \leq c_m \leq \frac{m}{j} t_j^{-(m+1)}. \quad (6.2)$$

The ratio of the upper bound to the lower bound for c_m is exactly $1 + m/((m+1)j)$. For positive integer m , all endpoints are rational.

Proof. Integrating [equation \(5.4\)](#) between 0 and t_j gives

$$0 \leq L_m - f_m(t_j) \leq \int_0^{t_j} t^m dt = \frac{t_j^{m+1}}{m+1}.$$

Since $f_m(t_j) = (j/m)t_j^{m+1}$, this proves [equation \(6.1\)](#). All quantities are positive, so taking reciprocals proves [equation \(6.2\)](#). The endpoint ratio follows by cancellation. When m is a positive integer, the factors defining t_j and the powers in these formulas are rational. $\square$

These intervals certify the constant without floating-point Gamma evaluation. Their relative gap tends to zero at an explicit rate. They do not by themselves certify how close a finite threshold $T_m(K)$ is to its asymptotic equivalent: that is a different error problem.

There is also an exact finite computation bound. If the backward recurrence has been computed only as far as stage k , then

$$k^m q_k \leq T_m(K) < k^m q_k + \sum_{r=1}^{k-1} r^m \quad (k \geq 2). \quad (6.3)$$

For integer m , this is a certified rational or integer enclosure of a particular threshold, independent of the asymptotic theorem.

6.2 An integral interpretation of the constant

Since $f_m(1) = 0$, [equation \(5.4\)](#) also gives

$$L_m = \int_0^1 t^m \left[C\left(\frac{my_m(t)}{t}\right) - \frac{my_m(t)}{t} \right] dt. \quad (6.4)$$

The term in brackets is a deterministic rounding residual. Formula [equation \(6.4\)](#) shows precisely how the successive increment regimes determine its weighted average. Replacing this residual by a fixed average would discard the dependence of its phase on the current quotient.

The full-trajectory theorem gives a related limit for the distribution of the losses. For a trajectory ending at $J = \tau_m(n)$, define a probability measure on $[0, 1]$ by

$$\mu_n = \sum_{k=2}^J \frac{x_{k-1} - x_k}{n} \delta_{k/J}. \quad (6.5)$$

Here δ_u denotes unit mass at u . The weights sum to one because $x_1 = n$ and $x_J = 0$.

Corollary 6.2 (Limiting distribution of rounding losses). *The measures μ_n converge weakly to the absolutely continuous probability measure with distribution function $1 - F_m(t)$ and density*

$$\rho_m(t) = c_m t^m \left[C\left(\frac{my_m(t)}{t}\right) - \frac{my_m(t)}{t} \right] \quad \text{for almost every } 0 < t < 1. \quad (6.6)$$

In particular, $0 \leq \rho_m(t) \leq c_m t^m$ and $\int_0^1 \rho_m(t) dt = 1$. The same conclusion holds under the smooth-weight hypothesis of [theorem 1.3](#).

Proof. The distribution function of [equation \(6.5\)](#) is $1 - X_n(t)$. The uniform convergence of the forward paths therefore proves convergence of these distribution functions to $1 - F_m$. The derivative formula in [equation \(5.4\)](#), including its integrable bound at zero, shows that the limit is absolutely continuous and gives [equation \(6.6\)](#). For smooth weights, use the forward-profile part of [theorem 1.3](#), proved below. $\square$

No randomness has been introduced in this statement. The measure records what fraction of a fixed initial value is lost at each normalized stage.

6.3 Monotonicity and endpoint behavior of c_m

The explicit constant permits useful parameter calculations. Let γ be Euler's constant and ζ the Riemann zeta function.

Proposition 6.3 (Dependence on m). *The function $m \mapsto c_m$ is strictly increasing on $(0, \infty)$. Moreover,*

$$\log c_m = -m \log m + (1 - \gamma)m + O(m^2), \quad m \downarrow 0, \quad (6.7)$$

$$\log c_m = \log m + \gamma + \sum_{r=2}^{\infty} \frac{\zeta(r)}{r(m+1)^{r-1}}, \quad m > 0, \quad (6.8)$$

$$c_m = e^\gamma m \left(1 + \frac{\pi^2}{12(m+1)} + O(m^{-2}) \right), \quad m \rightarrow \infty. \quad (6.9)$$

In particular, $c_m \rightarrow 1$ as $m \downarrow 0$, and c_m/m decreases strictly to e^γ as $m \rightarrow \infty$.

Proof. The convergent expansion

$$\log \Gamma(1 - z) = \gamma z + \sum_{r=2}^{\infty} \frac{\zeta(r)}{r} z^r, \quad |z| < 1,$$

is a standard Gamma identity [10]. Substitute $z = 1/(m+1)$ to obtain equation (6.8). Its positive terms decrease with m , which proves the assertion about c_m/m and the expansion equation (6.9). For small m , put $a = m/(m+1)$ and use $\log \Gamma(a) = -\log a - \gamma a + O(a^2)$; substitution in equation (1.2) gives equation (6.7).

For strict monotonicity of c_m itself, logarithmic differentiation gives

$$\frac{d}{dm} \log c_m = \frac{1}{m} + \log \Gamma(a) + (1 - a)\psi(a), \quad a = \frac{m}{m+1},$$

where ψ is the digamma function. Using its functional equation, this is

$$g(a) = \log \Gamma(1 + a) - \log a + (1 - a)\psi(1 + a).$$

We have $g(1) = 0$, and the standard trigamma series [10] gives

$$g'(a) = -\frac{1}{a} + (1 - a)\psi'(1 + a), \quad \psi'(1 + a) = \sum_{r=1}^{\infty} \frac{1}{(r + a)^2} < \int_a^{\infty} \frac{du}{u^2} = \frac{1}{a}.$$

Thus $g'(a) < 0$ for $0 < a < 1$ and consequently $g(a) > g(1) = 0$. $\square$

The limit $m \downarrow 0$ is singular for the original process. At $m = 0$, every modulus is 1, so a positive integer initial value never changes and has no finite extinction time. The fixed- m theorem does not justify interchanging the limits $m \downarrow 0$ and $n \rightarrow \infty$.

7 Smoothly varying moduli

This section proves theorem 1.3 in full. The useful information in equation (1.12) is the local relative increment of the modulus; no differentiability of an interpolation is required.

7.1 Elementary consequences of the ratio hypothesis

Lemma 7.1 (Regular variation and a summable tail). *Under [equation \(1.12\)](#), the weights tend to infinity and are eventually strictly increasing. For every compact interval $[\delta, M] \subset (0, \infty)$,*

$$\frac{w_{\lfloor Kt \rfloor}}{w_K} \longrightarrow t^m \quad \text{uniformly for } \delta \leq t \leq M. \quad (7.1)$$

For each $0 < \delta \leq 1$,

$$\frac{1}{Kw_K} \sum_{1 \leq j < \delta K} w_j \longrightarrow \frac{\delta^{m+1}}{m+1}. \quad (7.2)$$

Proof. The ratio hypothesis implies

$$\log(w_k/w_{k-1}) = \frac{m + o(1)}{k}. \quad (7.3)$$

In particular the ratios are eventually larger than one, and the divergence of the harmonic series implies $w_k \rightarrow \infty$. Summing [equation \(7.3\)](#) between K and $\lfloor Kt \rfloor$ proves [equation \(7.1\)](#). Uniformity follows because all relevant indices are at least a fixed positive multiple of K , while the sums of their reciprocals are uniformly bounded on $[\delta, M]$.

For the sum near zero, choose $0 < \varepsilon < m$. For all sufficiently large ℓ ,

$$\log(w_\ell/w_{\ell-1}) \geq \frac{m - \varepsilon}{\ell}.$$

It follows, for all sufficiently large $j \leq K$, that

$$\frac{w_j}{w_K} \leq \exp\left(- (m - \varepsilon) \sum_{\ell=j+1}^K \frac{1}{\ell}\right) \leq \left(\frac{j+1}{K+1}\right)^{m-\varepsilon}. \quad (7.4)$$

The finitely many omitted weights have contribution tending to zero after division by Kw_K . On $[\eta, \delta]$ for $\eta > 0$, the locally uniform limit in [equation \(7.1\)](#) gives the usual Riemann sum. The bound in [equation \(7.4\)](#) makes the normalized sum over $j < \eta K$ uniformly small as $\eta \downarrow 0$. Thus the full limit is $\int_0^\delta t^m dt$, proving [equation \(7.2\)](#). $\square$

The proof of this lemma is included to keep the main theorem independent of general regular-variation machinery. Its conclusion is the familiar regular variation of index m , obtained here from an explicit successive-ratio assumption.

7.2 The same backward profile

Define

$$q_K^{(K)} = 1, \quad q_{k-1}^{(K)} = \left\lceil \frac{w_k}{w_{k-1}} q_k^{(K)} \right\rceil, \quad T_w(K) = q_1^{(K)}. \quad (7.5)$$

The proof of [lemma 3.2](#) applies without change:

$$\tau_w(n) > K \iff n \geq T_w(K). \quad (7.6)$$

This remains valid even if finitely many early weights decrease. The normalization $w_1 = 1$ ensures that the integer initial values are exactly the admissible quotients at stage 1.

The exact error identity is now

$$0 \leq w_{k-1}q_{k-1} - w_k q_k < w_{k-1}. \quad (7.7)$$

It gives

$$w_k q_k \leq w_K + \sum_{j=k}^{K-1} w_j, \quad (7.8)$$

$$0 \leq T_w(K) - w_k q_k < \sum_{j=1}^{k-1} w_j \quad (k \geq 2). \quad (7.9)$$

For $k \geq \delta K$, eventual monotonicity bounds the right-hand side of [equation \(7.8\)](#) by $(K+1)w_K$. Together with [equation \(7.1\)](#), this gives a uniform bound on q_k/K .

Set $b_k = w_k/w_{k-1} - 1$. On these macroscopic intervals,

$$q_{k-1} - q_k = \lceil b_k q_k \rceil, \quad b_k = \frac{m + o(1)}{k} > 0.$$

Thus the increments are uniformly bounded and at least one. The compactness, positive-drift, and ceiling-discontinuity arguments of [section 4](#) apply verbatim. The limit equation is still $y' = -C(my/t)$, and its unique selected solution is y_m . Therefore

$$\frac{q_{\lfloor Kt \rfloor}^{(K)}}{K} \longrightarrow y_m(t) \quad \text{locally uniformly on } (0, 1]. \quad (7.10)$$

The finitely many weights before monotonicity begins are outside each fixed macroscopic interval once K is large.

7.3 Matching and inversion

Proof of [theorem 1.3](#). Set $k = \lfloor \delta K \rfloor$ in [equation \(7.9\)](#). By [equations \(7.1\)](#) and [\(7.10\)](#),

$$\frac{w_k q_k}{K w_K} \longrightarrow \delta^m y_m(\delta).$$

By [equation \(7.2\)](#), the normalized error sum tends to $\delta^{m+1}/(m+1)$. The same sandwich used for [theorem 1.1](#), followed by $\delta \downarrow 0$, gives $T_w(K)/(K w_K) \rightarrow L_m$.

Let $J = \tau_w(n)$. Since $w_k \rightarrow \infty$, extinction is finite; as $n \rightarrow \infty$, its index tends to infinity. Formula [equation \(7.6\)](#) gives

$$T_w(J-1) \leq n < T_w(J).$$

The ratio hypothesis implies $w_{J-1}/w_J \rightarrow 1$, so both bounds are asymptotic to $L_m J w_J$. This proves [equation \(1.13\)](#).

For the forward trajectory, put $Q_k = x_k/w_k$. Survival through $J-1$ but not J gives the exact sandwich

$$q_k^{(J-1)} \leq Q_k < q_k^{(J)} \quad (1 \leq k \leq J-1).$$

Equation [\(7.10\)](#) yields $Q_{\lfloor Jt \rfloor}/J \rightarrow y_m(t)$ locally uniformly in $(0, 1)$. Therefore

$$\frac{x_{\lfloor Jt \rfloor}}{n} = \frac{w_{\lfloor Jt \rfloor}}{w_J} \cdot \frac{Q_{\lfloor Jt \rfloor}/J}{n/(J w_J)} \longrightarrow \frac{t^m y_m(t)}{L_m} = F_m(t).$$

The endpoint values and the finite-mesh monotonicity argument from [theorem 1.2](#) again give uniform convergence on $[0, 1]$. $\square$

7.4 Examples and consequences

Corollary 7.2 (Logarithmic modifiers). *Let $m > 0$, $b \in \mathbb{R}$, and $A > 0$. Take $w_1 = 1$ and $w_k = Ak^m(\log k)^b$ for $k \geq 2$. Then*

$$\tau_w(n) \sim \left(\frac{c_m(m+1)^b}{A} \frac{n}{(\log n)^b} \right)^{1/(m+1)}. \quad (7.11)$$

The same leading result holds if the weights are changed at finitely many stages.

Proof. Logarithmic differences give

$$k \log(w_k/w_{k-1}) = m + \frac{b}{\log k} + o(1),$$

so [equation \(1.12\)](#) holds. The threshold theorem gives $AJ^{m+1}(\log J)^b \sim c_m n$. In particular $\log J \sim \log n/(m+1)$; substitution yields [equation \(7.11\)](#). A finite change leaves the ratio hypothesis and the asymptotic expression for w_K unchanged, so the same theorem applies. $\square$

One may similarly take $w_k = A(k + \sigma)^m$ for all sufficiently large k , where $k + \sigma > 0$. The stopping equivalent is $(c_m n/A)^{1/(m+1)}$. More generally, if $w_k = k^m \ell(k)$ satisfies [equation \(1.12\)](#), the clean inverse statement is

$$\tau_w(n)^{m+1} \ell(\tau_w(n)) \sim c_m n.$$

The profile theorem remains explicit even when this inverse has no elementary closed form.

Corollary 7.3 (Regular variation of the stopping index). *Under [equation \(1.12\)](#), for every $\lambda > 0$,*

$$\frac{\tau_w(\lfloor \lambda n \rfloor)}{\tau_w(n)} \longrightarrow \lambda^{1/(m+1)}.$$

Proof. The increasing function $K \mapsto L_m K w_K$ is eventually regularly varying with positive index $m+1$, by [equation \(7.1\)](#). For any constants $0 < a < \lambda^{1/(m+1)} < b$, its ratios at $\lfloor aK \rfloor$ and $\lfloor bK \rfloor$ tend respectively to $a^{m+1} < \lambda$ and $b^{m+1} > \lambda$. Apply these inequalities at $K = \tau_w(n)$ and use [equation \(1.13\)](#) and monotonicity of the stopping index. Let a and b approach $\lambda^{1/(m+1)}$. $\square$

8 Why the microscopic schedule matters

It is tempting to regard $w_k \sim Ak^m$ as all the information needed to determine extinction. The following exact construction shows why the local ratio condition is substantive.

Proposition 8.1 (Repeating each modulus). *Fix an integer $r \geq 2$ and put*

$$w_k = \lceil k/r \rceil^m. \quad (8.1)$$

For every integer $n > 0$ and terminal stage $K \geq 1$,

$$\tau_w(n) = r(\tau_m(n) - 1) + 1, \quad (8.2)$$

$$T_w(K) = T_m(\lceil K/r \rceil). \quad (8.3)$$

Although $w_k \sim (k/r)^m$ and the weights are regularly varying with index m ,

$$\frac{T_w(K)}{K w_K} \longrightarrow \frac{L_m}{r}. \quad (8.4)$$

Proof. The operation $R_a(x) = a \lfloor x/a \rfloor$ is idempotent: $R_a(R_a(x)) = R_a(x)$. Each modulus j^m occurs at the stages $(j-1)r+1, \dots, jr$. Only its first occurrence can change the current value. At the initial block the modulus is 1, so it leaves the integer n unchanged. If the unblocked process first vanishes when modulus J^m is used, the blocked process first vanishes at stage $(J-1)r+1$. This proves [equation \(8.2\)](#).

Survival through stage K of the blocked process is equivalent to survival through modulus $\lceil K/r \rceil^m$ of the unblocked process. Duality gives [equation \(8.3\)](#). Apply [equation \(1.4\)](#) and $w_K = \lceil K/r \rceil^m$ to obtain

$$\frac{T_w(K)}{Kw_K} \sim L_m \frac{\lceil K/r \rceil}{K} \longrightarrow \frac{L_m}{r}.$$

□

For $n = 0$, both stopping times are 1; the displayed positive- n formula need not be used. In the block example the quantity $k(w_k/w_{k-1} - 1)$ is zero inside each block and approaches rm at its jumps. It therefore has no limit equal to m .

For comparison, choose smooth weights $v_1 = 1$ and $v_k = (k/r)^m$ for $k \geq 2$. They have the same leading equivalent as [equation \(8.1\)](#), but satisfy the local ratio hypothesis. Their stopping equivalent is

$$\tau_v(n) \sim (c_m r^m n)^{1/(m+1)},$$

whereas [equation \(8.2\)](#) gives $\tau_w(n) \sim r(c_m n)^{1/(m+1)}$. The ratio tends to $r^{1/(m+1)}$. Thus even the leading stopping constant distinguishes the two schedules.

This example supplies a precise research boundary. A theorem for all regularly varying moduli must include additional information about how their growth is distributed among individual rounding steps.

9 Exact computation and numerical evidence

The accompanying program serves two purposes: it checks that independent finite implementations agree with the definitions, and it illustrates the asymptotic theorems. All floor and ceiling decisions in the reported checks use exact integers. Special-function evaluations and plots are separately identified as numerical diagnostics.

9.1 Integer and rational exponents

For integer $m \geq 1$, the backward recurrence has the purely integer form

$$q_{k-1} = \left\lfloor \frac{k^m q_k + (k-1)^m - 1}{(k-1)^m} \right\rfloor. \quad (9.1)$$

It computes $T_m(K)$ in $K-1$ arithmetic steps and uses only one current quotient if no profile is retained. These are counts of arithmetic operations, not a bit-complexity bound. For fixed m , all integer values involved have $O_m(\log K)$ bits by [equation \(3.9\)](#).

Exact computation is also possible for rational $m = p/q > 0$, written in lowest terms. Given the integer quotient $v = q_k$, the next backward quotient is the least integer u satisfying

$$u^q (k-1)^p \geq v^q k^p. \quad (9.2)$$

This is an integer-root problem. Let $N = v^q k^p$, $D = (k-1)^p$, and let r be the integer q th root rounded down of $\lfloor N/D \rfloor$. The desired ceiling is r if $r^q D = N$, and $r+1$ otherwise. The final comparison is exact, including the cases where the algebraic expression is an integer.

Independently, the forward quotients satisfy

$$Q_1 = n, \quad Q_k = \left\lfloor \left(\frac{k-1}{k} \right)^m Q_{k-1} \right\rfloor. \quad (9.3)$$

For $m = p/q$, this is the greatest integer u such that $u^q k^p \leq Q_{k-1}^q (k-1)^p$. Thus rational exponents can be checked without ever representing an irrational intermediate x_k .

9.2 Independent checks

The recorded verification performs the following mathematical checks:

- For $m = 1, 2, 3$ and every $0 \leq n \leq 2000$, direct simulation of the original x_k recurrence agrees with the quotient recurrence and with the exact backward-threshold interval for the stopping time.
- For $m = 1/2, 2/3, 3/2, 5/3$ and $0 \leq n \leq 200$, the algebraic integer-root implementation of the forward process agrees with the backward survival intervals.
- All displayed initial terms in the inspected OEIS entries agree: 79 terms for A073047 with offset 1, and 105 terms each for A082527 and A082528 with offset 0.
- For block length $r = 3$ and $m = 1, 2, 3$, the exact threshold identity is checked through $K = 200$, and the stopping identity through $n = 2000$. The zero input is checked separately.
- The finite-product constant enclosures are constructed as exact fractions for $m = 1, 2, 3, 5$ and $j = 10, 100, 1000$, and their stated relative width is checked exactly.

These checks all passed. Their bounds and results are recorded in `verification.json`. They audit the finite definitions and calculations; the infinite statements rest on the preceding proofs.

9.3 Threshold convergence

Table 1 shows both the correct scale and the irregular finite errors. For example, for $m = 2$ the normalized error at $K = 10000$ is larger than at $K = 1000$. Neither monotonic convergence nor a particular rate follows from the theorem or from these samples.

9.4 The predicted shapes

Figure 2 directly illustrates the compact convergence proved in [proposition 4.3](#). Figure 3 uses actual forward trajectories, which were not used to construct the limiting profile. For the cubic process, the exact stopping indices for initial values $10^3, 10^6, 10^9$ are respectively 9, 51, 287. The step curves converge to the continuous weighted profile F_3 .

9.5 Certified constant intervals

The exact rational numerators and denominators underlying Table 2 are included in

`data/constant_certificates.json`

The table's enclosure is provided by the theorem and exact endpoint arithmetic. A numerical comparison of a Gamma evaluation with those intervals is an additional consistency check, not the basis of their certification.

m	K	Exact $T_m(K)$	$c_m T_m(K)/K^{m+1}$
$\frac{1}{2}$	10	17	1.1785908081
$\frac{1}{2}$	100	470	1.0304145588
$\frac{1}{2}$	1000	14440	1.0011088982
$\frac{1}{2}$	10000	456247	1.0002628749
1	10	34	1.0681415022
1	100	3234	1.0159910642
1	1000	318570	1.0008171717
1	10000	31833630	1.0000829815
2	10	244	1.2116837876
2	100	205688	1.0214295693
2	1000	201388916	1.0000806743
2	10000	201457639236	1.0004219482
3	10	1816	1.2284918534
3	100	15338896	1.0376491616
3	1000	148522784144	1.0047303434
3	10000	1478758856906056	1.0003541899
5	10	141440	1.4628989077
5	100	103280328768	1.0682174784
5	1000	97237370572647616	1.0057158031
5	10000	96759789128216456247328	1.0007762290

Table 1: Exact backward thresholds and numerical normalizations. The last column tends to 1 by [theorem 1.1](#); its displayed values use high-precision evaluation of c_m . The full data file also includes $K = 100000$. The exponent $m = 1/2$ uses exact integer square roots.

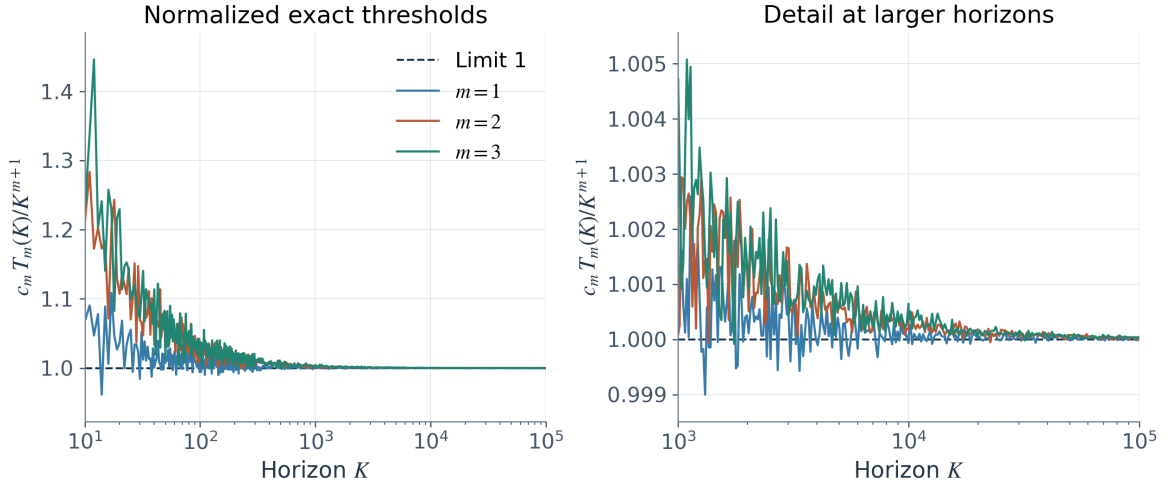


Figure 1: Normalized thresholds for $m = 1, 2, 3$, calculated by the exact backward recurrence. The Gamma constants are used only in the normalization. The irregular excursions illustrate why estimating a discrete error law requires more than a smooth fitted curve.

m	Lower endpoint	$m\Gamma(m/(m+1))^{m+1}$	Upper endpoint
1	3.1408077460	3.1415926536	3.1423781500
2	4.9642631432	4.9659171624	4.9675726521
3	6.7622883375	6.7648229810	6.7673600538
5	10.3385883285	10.3428938609	10.3472038189

Table 2: Finite-product enclosures from [proposition 6.1](#) at $j = 1000$. The outer decimal endpoints are rounded outward by exact integer arithmetic, so the displayed intervals retain the enclosure. The middle column is a numerical evaluation of the exact Gamma formula.

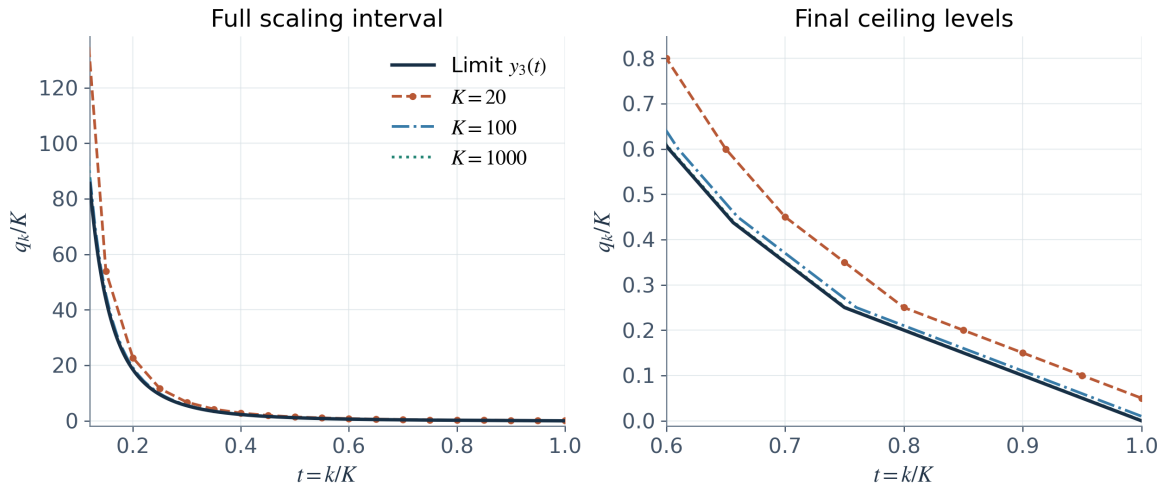


Figure 2: The cubic backward quotient profile. Discrete arrays, divided by their terminal stage K , approach the explicit piecewise linear function y_3 . The plot stays away from $t = 0$, where y_3 diverges; the proof handles that endpoint through weighted tail bounds.

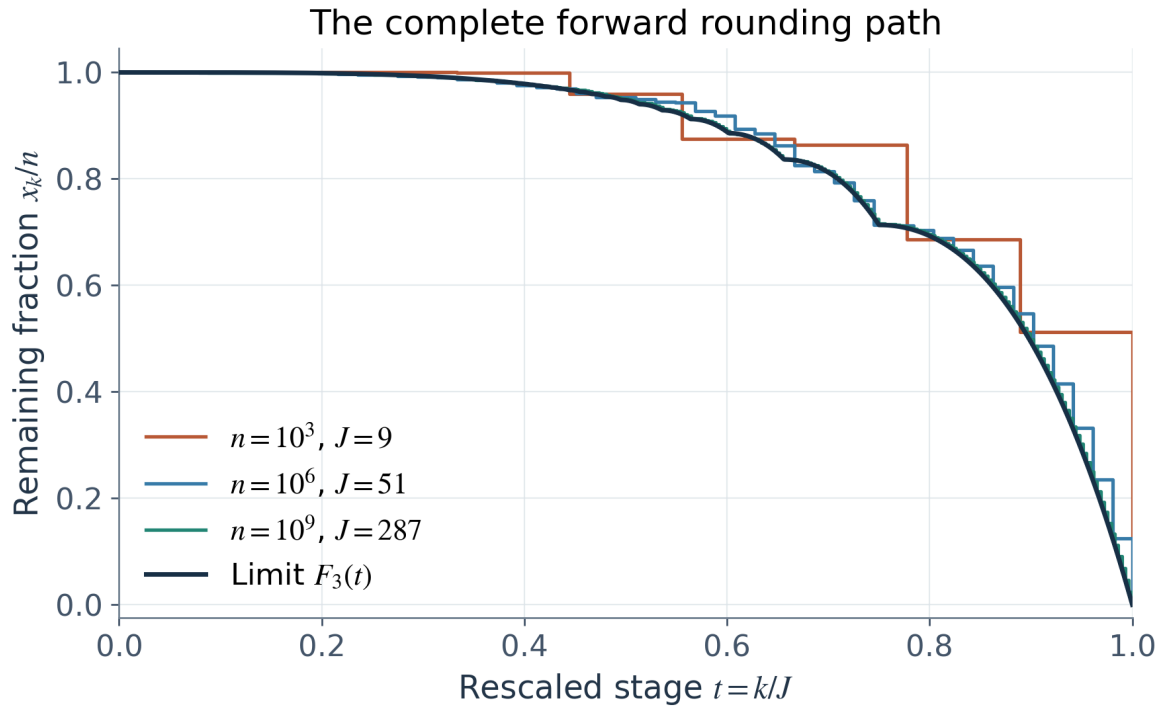


Figure 3: The fraction of the initial value remaining in three cubic forward trajectories, with time divided by the exact stopping index. The limiting curve is $F_3(t) = c_3 t^3 y_3(t)$. Its continuous value at $t = 0$ is 1, despite the divergence of the unweighted quotient profile.

9.6 Reproduction and formalization

The archive includes the LaTeX source, this PDF, the figures, the Python program, its dependency versions, and recorded data. From the archive's root directory, the full calculation is reproduced by

```
python reproduce.py
```

The exact checks alone require only the Python standard library:

```
python reproduce.py --exact-only --out replay_exact
```

The full replay additionally uses `mpmath` and `matplotlib`; it requires no network access. Compile `article.tex` twice with `pdflatex` to rebuild the PDF. The README gives query examples, output descriptions, and precision conventions.

A formalization can be divided at natural mathematical interfaces. The finite recurrence and survival duality need only integer order, floor/ceiling inequalities, and finite induction. Rational exponents admit an exact integer-root specification. The analytic part requires compactness of uniformly Lipschitz functions on compact intervals, dominated convergence, the Gamma functional equation, and the beta integral. The proof isolates the delicate points as reusable lemmas: the positive drift that selects the terminal solution, the finite number of ceiling discontinuities on compact intervals, and the final weighted tail estimate.

No proof-assistant development accompanies this report. The executable checks are finite computations and are not described as a formal verification of the real-parameter theorem.

[write] Added 3 October 2026, batch 85. This section describes the delivered archive. In the repository the files keep their bytes but not all their places: the program is `code/reproduce.py`, `requirements.txt` and the recorded `verification.json` are in `data/`, the delivered `README.txt` is replaced by the report's `README.md`, and the delivered 25-page PDF was not shipped (the repository's `article.pdf` is a rebuild of this source with the [write] notes; the delivered one survives in the arrival commit). The program reads `data/oeis_prefixes.json` relative to its own directory and, without `--out`, writes into that directory, so it must not be run in place: run from `code/` it would not find its fixture file. The command above was printed in the delivered PDF with en dashes for its double hyphens (a typewriter-font ligature); the writing step set them as two hyphens, the only change to the delivered text outside the labels, the notes and the preamble. The report's `README.md` rebuilds the delivered layout on a scratch copy and replays there. At intake (3 October 2026) the full replay on such a copy passed in 92.5s; the seven CSV files were byte-identical to the recorded ones, the two JSON and two T_EX data files differed only by CRLF line ends (Python text mode on Windows), `verification.json` only in its runtime and Python-version fields, and the three PNG figures only in rendering. The OEIS terms in `data/oeis_prefixes.json` are OEIS content, licensed by The OEIS Foundation under CC BY-SA 4.0; they are not covered by the repository's MIT-0 licence. No Lean or Rocq development in ProveIt treats this subject.

10 Further research questions

The leading constant is only the first part of the rounding problem. The following questions separate plausible next steps from conclusions already proved. They are proposed research problems; except for the historical linear question, their open status refers to this report and the sources inspected, not to an exhaustive classification of the literature.

Research question 1 (Quantitative threshold errors). For fixed $m > 0$, find a usable rate in

$$T_m(K) = L_m K^{m+1} + E_m(K).$$

Which exponents $\delta_m > 0$ admit $E_m(K) = O_m(K^{m+1-\delta_m})$? Can an explicit bound be obtained uniformly for m in a compact subinterval of $(0, \infty)$?

The proof identifies the correct place to work: make the crossing times of the discrete quotient increments quantitative, then balance their accumulated errors against the omitted small-time tail. The classical $m = 1$ estimate in [equation \(2.1\)](#) supplies a benchmark. Compactness alone contains no numerical rate.

Research question 2 (Bounded stopping-time error). For which $m > 0$, if any, does

$$\tau_m(n) = (c_m n)^{1/(m+1)} + O_m(1)$$

hold? In particular, resolve the stronger bounded-error claim for $m = 1$ recorded in A073047.

For pure powers, this is equivalent to $T_m(K) = L_m K^{m+1} + O_m(K^m)$. One direction follows by inverting the two neighboring threshold bounds. For the other, evaluate the stopping-time formula at the first integer of each survival interval, where the stopping index is $K + 1$, and expand the resulting power. The two equivalent formulations indicate that an error term at the scale of a single threshold gap is required.

Research question 3 (Signs and fluctuation scales). Does $T_m(K) - L_m K^{m+1}$ have an eventual sign? What are the natural scales of its upper and lower envelopes? Does any centered and scaled version have a limiting distribution along integer horizons?

The selected tabulated horizons have positive errors, while the denser plotted data include both signs. Neither observation establishes an eventual-sign theorem. The nonmonotone normalized errors show that a simple monotonicity argument would need additional structure. Any probabilistic terminology here would describe a distribution over sampled horizons, not randomness in the recurrence.

Research question 4 (A theory of microscopic schedules). Classify the limits for regularly varying weights when $k(w_k/w_{k-1} - 1)$ oscillates, has repeated zero values, or follows a periodic pattern. Which information beyond the regular-variation index determines the extinction constant and the limiting profile?

The block construction gives a completely solved test family. A possible next class consists of a fixed periodic pattern of small relative increments. The ceiling acts before any averaging, so an effective drift would have to be derived from the individual phases of that pattern. Establishing such an averaging theorem is a substantive extension; it is not a consequence of [theorem 1.3](#).

Research question 5 (Stability under sparse exceptional steps). How many exceptional ratios can be allowed without changing L_m ? For example, suppose the local ratio condition holds outside a set of indices of density zero. Which bounds on the exceptional increments suffice, and can a sparse set of very large jumps change the constant?

Finite modifications are already covered. Passing from finitely many exceptions to a sparse infinite set requires controlling their combined effect on the quotient increments. Density alone need not control the size of those effects.

Research question 6 (Moving exponents and singular regimes). Describe joint limits with $m = m(n) \downarrow 0$ or $m = m(n) \rightarrow \infty$. Determine ranges in which the fixed-parameter equivalent remains uniform and identify any transition scales.

The discontinuity at $m = 0$ makes the first question particularly interesting: positive integer values never vanish when every weight is exactly 1. The explicit expansions in [proposition 6.3](#) describe the constants, but they do not control the speed of convergence of the underlying discrete process.

Research question 7 (Rates for the full trajectory and loss distribution). Obtain explicit bounds for

$$\sup_{0 \leq t \leq 1} |X_n(t) - F_m(t)|$$

and for convergence of the loss measures in [corollary 6.2](#). Can one derive asymptotics for individual moments or quantiles of the loss-time distribution with a controlled error?

The limiting density is explicit on each increment regime. It permits certified computation of limiting observables using finite sums and an integrable tail. Rates for the finite process again depend on discrete crossing errors rather than solely on the Gamma constant.

Research question 8 (Sieve and game interpretations for higher powers). Is there a natural sieve or solitaire model whose survival thresholds are $T_m(K)$ for integer $m \geq 2$? Can such a model explain the Gamma product combinatorially or connect the threshold gaps to an existing family of integer sequences?

The ordinary case has the Tchoukaillon interpretation. A higher-power model could expose monotonicity, congruence, or conservation properties not apparent in the floor recurrence. Any proposed model should first be proved equivalent to the exact backward recurrence, including its initial conditions.

Research question 9 (Faster exact algorithms and formal verification). Can long consecutive ranges with a constant quotient increment be skipped in a provably correct exact algorithm? Can the general real-parameter theorem be formalized with a reusable library lemma for monotone discontinuous drift limits?

The present algorithm takes one arithmetic step per stage. The limiting piecewise linear structure suggests acceleration, but the exact locations of discrete transitions must be certified before steps can be skipped. For formalization, a useful first milestone is the exact survival duality and rational-exponent algorithm, followed by the compactness and endpoint-selection lemmas.

10.1 A practical order for continuing the work

The most direct continuation is a quantitative crossing-time analysis for the pure power family. In parallel, a periodic-ratio model can test how the constant changes when the local smoothness condition is relaxed. These two directions address the principal limitations of the present theorem: the absence of a finite error bound, and the dependence on the microscopic schedule.

A Proof dependencies and source audit

A.1 Logical dependencies

The extinction proof proceeds through five independently checkable steps:

1. Exact stagewise survival duality gives the backward ceiling recurrence and the inverse threshold relation.

2. The integer increment bound and weighted rounding-error bound give compactness and force positive terminal drift.
3. Strict monotonicity of $my(t)/t$ controls the ceiling discontinuities, yielding a rigorous limit equation.
4. Solving its increment regimes gives the finite product; the beta integral evaluates its asymptotic constant.
5. A telescoping estimate controls the entire omitted initial part, allowing the two limits to be taken in the correct order.

The smooth-weight theorem reuses these steps after proving the local ratio and tail-sum consequences of its hypothesis. The forward-path theorem follows from a second use of the exact stagewise duality. No numerical estimate is an input to any of these proofs.

Two independent mathematical reviews checked the ceiling passage, initial-value convention, smooth-weight tail, inversion, profile endpoint argument, and block counterexample. The exact finite verification was generated by a separate implementation. These reviews improve confidence but are not a substitute for peer review or formal verification.

A.2 Sources reviewed and scope of the novelty statement

The principal source records are the OEIS definitions and conjectures for A073047, A082527, and A082528; the classical threshold entry A002491; Erdős and Jabotinsky’s original sieve paper; the Broline–Loeb preprint and the criticism attributed to Knuth in the threshold entry; Brown’s exposition; Lehericy’s 2023 proof for a different generalized rounding rule; and Li’s 2026 Tchoukaillon preprint. The bibliography provides direct links.

Targeted searches combined the exact sequence identifiers with power-rounding, asymptotic, Gamma, generalized sieve, and the classical authors’ names. Searches within ProveIt used the exact sequence identifiers and Tchoukaillon-related terms. No matching real-power theorem or formula for c_m was found in that review. This statement is intentionally bounded by the sources and search scope. A conjectural label remaining on an OEIS entry does not prove that no proof exists elsewhere.

The result should therefore be described precisely: this report supplies a complete proof of the real-parameter conjecture as recorded in A082528, identifies the square and cube constants, and proves the stated extensions. It makes no claim to have newly discovered the linear π law or the general increment-regime method. The numerical and symbolic evidence in the archive is reproducible and separately labeled.

B Suggested OEIS update text

The following concise mathematical statements are suitable for review before any submission. They have not been posted to OEIS.

[write] Added 3 October 2026, batch 85. This appendix is the manuscript’s *draft* of OEIS update text, printed as delivered; it is not posted. Nothing has been submitted to OEIS: the package’s `VERIFICATION_NOTES.txt` records that no OEIS updates were submitted, and the intake submitted nothing either. On 3 October 2026 A082528 still recorded the real- m law as a conjecture. Whether any of this text is ever proposed is a decision for a human editor after review of the proof; the repository does not submit it.

For A082527. The conjectured asymptotic constant is

$$c = 2\Gamma(2/3)^3 = 4.965917162430455473\dots$$

Thus $a(n) \sim (2\Gamma(2/3)^3 n)^{1/3}$. This is the $m = 2$ case of the power-rounding extinction theorem.

For A082528. For every fixed real $m > 0$, if $x_1 = n$ and $x_k = k^m \lfloor x_{k-1}/k^m \rfloor$, the first zero satisfies

$$a(n, m) \sim \left(m\Gamma\left(\frac{m}{m+1}\right)^{m+1} n \right)^{1/(m+1)}.$$

In particular, the cubic constant is $3\Gamma(3/4)^4 = 6.764822981008760234\dots$. A proof is given by the backward survival thresholds and their piecewise linear scaling limit.

For the linear cross-reference. The $m = 1$ specialization is the classical leading equivalent $a(n) \sim \sqrt{\pi n}$. It does not establish the stronger $O(1)$ error term in A073047.

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